Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **React-based interfaces**, **API-driven platforms**, and AI-enabled workflows, with experience in prototypes, testing, cloud-native delivery, and collaborative problem-solving for mission-focused applications.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant application services for 500+ users, reducing latency from **1.2s to 350ms** through SQL optimization, indexed access patterns, and backend improvements across cloud-backed user workflows.
- Built automated validation and ETL checks for 5,000+ monthly jobs, reducing production failures and accelerating issue resolution through earlier detection of regressions, runtime anomalies, and data-quality drift.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed full-stack React and Flask features handling 10,000+ daily requests, translating product requirements into responsive interfaces and maintainable APIs while improving usability, testing discipline, and release confidence.
- Built AI-assisted grading workflows with multimodal models, OCR, and secure persistence, reducing feedback turnaround by **60%** while supporting feature-rich production operations in a fast-moving startup environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python experimentation pipelines across 100,000+ records, enabling reproducible evaluation and proof-of-concept iteration while improving debugging quality and communication of model-driven system behavior.
- Produced technical reports and targeted validation scenarios translating findings into actionable improvements, supporting stronger design reviews, clearer handoffs, and better engineering decisions across collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting distributed integrations, structured delivery, and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence during collaborative release cycles.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based application with WebSockets and REST APIs, supporting 2M+ synchronized records and six core workflows through modular services, responsive interfaces, and maintainable frontend-backend integration.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift during iterative delivery and demonstrations.

Skills

Languages: JavaScript/TypeScript, Python, Java, C++, SQL, Bash
Frontend: React, Next.js, Angular.js, responsive UI development, Tailwind CSS familiarity, component-driven workflows
Backend & APIs: Node.js, Flask, FastAPI, REST APIs, microservices, WebSockets, system integration
Cloud & Delivery: Docker, Kubernetes, AWS, Azure, Git/GitHub, Jenkins, CI/CD, testing, proof-of-concept development

Projects

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an AI-powered Next.js SaaS platform for image editing and search, integrating scalable storage, responsive interfaces, and premium billing while improving conversion by **18%** through polished experiences.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with **C++** logic, React controls, and Gantt visualization, improving algorithm debugging efficiency by **20%** across simulated workloads.

FinNexus Code | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a secure fintech platform supporting 50,000+ users and 1M+ transactions, combining **Next.js**, TypeScript, Kubernetes, and real-time updates to demonstrate scalable mission-critical application patterns.